

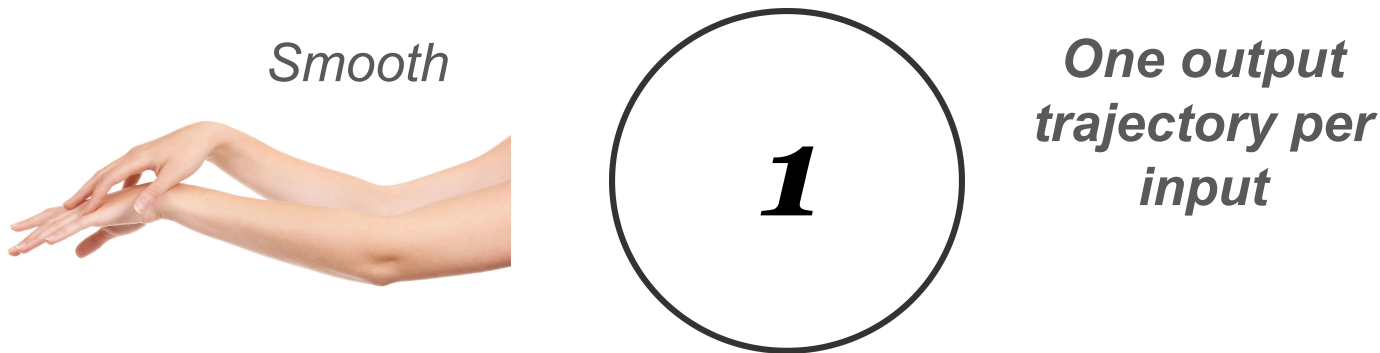
Understanding Neural Stochastic Differential Equations

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Looking Back

Neural ODE's

- Neural ODE's include *GRU-ODE*, *ODE-RNN*, and *ODE-LSTM*.
- They are models that **assume the system evolves in a smooth, fully predictable way over time.**
 - These **learn only the deterministic part of the dynamics** and therefore **produce a single, fixed trajectory for each input.**



Neural ODE's: Football Example

- Modeling the ideal **path of a ball** during a training drill with no defenders and **no disturbances**.
- The movement **depends only on the forces applied**, so the model can treat the motion as a clean, **deterministic system**.



Looking Forward

Neural SDE's: Adding Randomness

The Neural SDE (Stochastic Differential Equation) models are like Neural ODEs but add randomness to the dynamics. The model consists of **two neural networks**:

- **Drift Network**: determines the deterministic part of the change (*like standard Neural ODE*).
- **Diffusion Network**: determines the random (stochastic) part of the change.

Neural SDE's: Output

- Instead of giving just one predicted path (like Neural ODE's), Neural SDEs consider that the system can be unpredictable. So they **don't give one path**.
- Therefore, for the same starting point, the model can produce **many possible paths**, showing a **range of outcomes**.

Example: When rolling dice each roll is different, but all rolls follow certain rules.



***Output:**
Range of
Possibilities*

Neural SDE's: Zooming In

Key Question: How do Drift and Diffusion Networks work together to obtain the output range of possibilities?

Ingredients:

1. Time (measured in steps)
2. Drift Neural Network - outputs deterministic change (smooth movement at time step)
3. Diffusion Neural Network - outputs stochastic change (how much random movement at time step)
4. Random Noise - a **vector of possibilities** in terms of random variables

Formula:

- $\text{NextState} = \text{DeterministicChange@CurrentState} + (\text{StochasticChange@CurrentState} \times \text{RandomNoise@CurrentState})$

Neural SDE's: Variational Inference

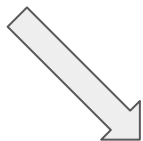
To make the range of output paths (predicted possibilities) realistic, **variational inference is used in training.**

- In a standard Neural ODE, one might use MSE (mean squared error) to measure how far predicted trajectory is from the observed trajectory.
- In Neural SDE, variational inference replaces MSE in the sense that it **measures how well the predicted distribution matches the observed data, rather than just comparing single points.**

So instead of penalizing distance from one trajectory it penalizes **how unlikely the observed data is to fall in the predicted distribution (possible outcomes).**

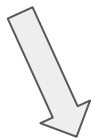
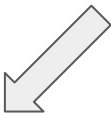
MSE (NODE)

Real Values =
(1.74, 1.64, 1.54)



Predicted Values =
(1.34, 1.24, 1.14)

MSE = 0.4

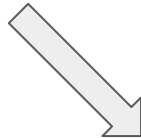


Update Model
Parameters

vs

Variational Inference (NSDE)

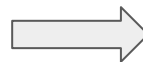
Real Values =
(1.74, 1.64, 1.54)



Predicted Normal
Distributions = ($x_1 \sim N(1.7, 0.05^2)$,
 $x_2 \sim N(1.65, 0.04^2)$,
 $x_3 \sim N(1.55, 0.03^2)$)



VI Likelihoods =
 $P(1.74|x_1)$,
 $P(1.64|x_2)$,
 $P(1.54|x_3)$



Update Model
Parameters

Presenting NSDE for COVID

Next Week Task Pitch

- Apply **Neural SDE model** to **Lynx-Hare Dataset**
- **Compare** to ODE-RNN performance
- **Look into ways** to improve model

